The Australian National University Semester 1, 2026
School of Computing COMP4670/8600
Conveners: Lexing Xie & Jing Jiang Weight: 20% - graded out of 100 points - See EdStem

Assignment 1: Bayesian, Exponential Family, Bias and Variance

Foreword and motivation Before delving into the actual questions, let's briefly explain their topics and goals – clarify their connections to our lecturer and tutorial materials, and what you can expect to learn from the assignment.

Questions need not be attempted in the order they are presented. Please feel free to work on whichever question for which you have some ideas on how to proceed. After working through all three questions you will hopefully feel more comfortable with linear regression in different forms and noise models, which forms the foundation of SML.

Question 0 considers machine learning methods that are non-Bayesian vs. Bayesian in nature. This warm-up question goes back to the basics and think about MLE and MAP estimates.

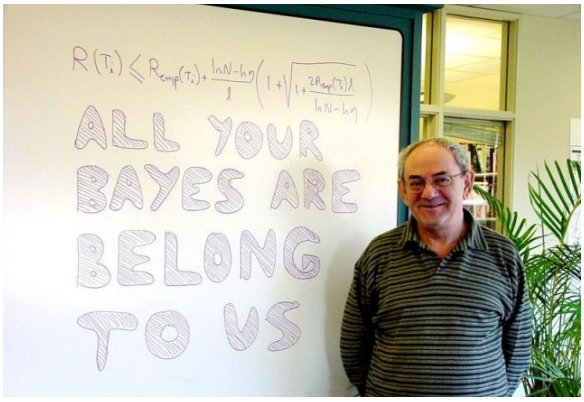

Figure 1: Vladimir Vapnik, shown with writings on the whiteboard being a play on the 1990s internet meme "All your base are belong to us" and a bound from statistical learning theory. Photo borrowed from `http://yann.lecun.com/ex/fun/index.html#allyourbayes`, date and location unknown.

Question 1 delves into Section 3.2 (page 149, shown below[1]) of the PRML book – it has obtained a reputation among previous SML students as being difficult to understand. Perceived barriers can include: the notion of *different datasets*, sampled i.i.d from the same training (and test!) distribution; the idea of taking *expectations* w.r.t. these datasets; and critically distinguishing a few seemingly similar quantities, such as underlying model, ground truth, and predictions.

Question 2 asks you to think about exponential family distributions – a general class of distributions with many nice properties, with well known members such as the normal distribution.

$$\text{expected loss} = (\text{bias})^2 + \text{variance} + \text{noise} \tag{3.41}$$

where

$$(\text{bias})^2 = \int \{\mathbb{E}_{\mathcal{D}}[y(\mathbf{x};\mathcal{D})] - h(\mathbf{x})\}^2 p(\mathbf{x})\,\mathrm{d}\mathbf{x} \tag{3.42}$$

$$\text{variance} = \int \mathbb{E}_{\mathcal{D}}\left[\{y(\mathbf{x};\mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\mathbf{x};\mathcal{D})]\}^2\right] p(\mathbf{x})\,\mathrm{d}\mathbf{x} \tag{3.43}$$

$$\text{noise} = \int \{h(\mathbf{x}) - t\}^2 p(\mathbf{x}, t)\,\mathrm{d}\mathbf{x}\,\mathrm{d}t \tag{3.44}$$

[1]Note that there is a typo in the pdf version of PRML book (but not the latest printed version) – (3.44) should have a double integral sign, rather than a single integral sign that does not correspond to the $dx\ dt$ at the end.

Section 1: Warm-up: Bayesian Thinking (10 Points)

Here is a brief refresher of applying *Bayes' Rule* to models and data in ML. Given a training dataset $\mathcal{D}$ and a model θ, the posterior distribution is defined as

$$\underbrace{\Pr(\theta \mid \mathcal{D})}_{\text{“posterior”}} = \frac{\overbrace{\Pr(\mathcal{D} \mid \theta)}^{\text{“likelihood”}} \cdot \overbrace{\Pr(\theta)}^{\text{“prior”}}}{\underbrace{\Pr(\mathcal{D})}_{\text{“data”}}} \propto \Pr(\mathcal{D} \mid \theta) \cdot \Pr(\theta). \tag{1.1}$$

On the right-hand side, $\Pr(\mathcal{D})$ is ignored and interpreted as a normalization constant for the posterior. Bayes' Rule incorporates *prior knowledge* into a model and we can consider using maximizing the *posterior* rather than the *likelihood* in parameter fitting.

Finding the MAP estimate consists in solving the optimization problem

$$\max_{\theta \in \Theta} \Pr(\theta \mid \mathcal{D}), \tag{1.2}$$

where Θ is the set of all possible values for θ. When $\Pr(\mathcal{D})$ is constant, it should be easy to convince ourselves that

$$\arg\max_{\theta} \Pr(\theta \mid \mathcal{D}) = \arg\max_{\theta} \Pr(\mathcal{D} \mid \theta) \Pr(\theta) \tag{1.3}$$

Question 1.1: Bayesian estimate for Normal distributions (2 Points)

Let $x_{1:n} = \{x_1, x_2, \ldots, x_n\}$ be a set of samples drawn from an unknown normal distribution $\mathcal{N}(x; \mu, \sigma^2)$. We derived in the lectures that the maximum likelihood estimate of the mean is the sample mean:

$$\hat{\mu}_{ML} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

and that

$$\hat{\sigma}^2_{ML} = \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{\mu}_{ML})^2$$

Now assume we have a Gaussian prior on the mean, i.e. $\mu \sim \mathcal{N}(\mu; \eta, \beta^{-1})$. Write the posterior probability $p(\mu | x_{1:n})$, and derive the MAP (maximum a posteriori) estimate μ^* for the mean.

Question 1.2: MLE vs MAP (2 Points)

Is there one or more prior distributions that will make MLE and MAP estimates equal (for estimating Gaussians, and for estimating all other distributions)? If yes, describe such a prior distributions.

Assumption: For the rest of this question, let's assume: for problem $\max_{\theta \in \Theta} \Pr(\theta \mid \mathcal{D})$, we assume there exists no method for finding or numerically approximating (with high enough precision) its optimal solution. However, we have access to $N > 0$ i.i.d. samples $\{\theta_i \sim \Pr(\cdot \mid \mathcal{D})\}_{i=1}^{N}$ and their corresponding probabilities $\Pr(\theta_i \mid \mathcal{D}), \forall i = 1, \ldots, N$.

You might notice from the Bayesian Regression lecture that, by the properties of Gaussian distributions, we can find the MAP estimator for θ if sampling from the posterior $\Pr(\theta \mid \mathcal{D})$ is possible. That is, assuming that (i) the posterior is Gaussian $(*)$ and (ii) we have N i.i.d. samples from $\Pr(\theta \mid \mathcal{D})$, we have

$$\arg\max_{\theta} \Pr(\theta \mid \mathcal{D}) \overset{(*)}{=} \mathbb{E}_{\theta|\mathcal{D}}[\theta] \approx \frac{1}{N} \sum_{i=1}^{N} \theta_i. \tag{1.4}$$

The sample mean on the right-hand side provides a convenient method of approximating the "best" model.

Now consider $\Pr(\theta \mid \mathcal{D})$ is not Gaussian. A tantalizing question emerges: When is it suitable to approximate the MAP estimator using a sample mean?

Question 1.3: Approximating the MAP (4 Points)

Consider a posterior distribution $\Pr(\theta \mid \mathcal{D})$ supported on Θ and N i.i.d. samples $\{\theta_i \in \Theta\}_{i=1}^N$ from it. Define the sample mean as $\frac{1}{N}\sum_{i=1}^N \theta_i$.

1. Describe key properties of posterior distributions, under which it is appropriate to use $\frac{1}{N}\sum_{i=1}^N \theta_i$ to approximate $\arg\max_{\theta\in\Theta} \Pr(\theta \mid \mathcal{D})$?

2. For what kind of posterior distribution for which $\frac{1}{N}\sum_{i=1}^N \theta_i$ would be a very poor approximate of $\arg\max_{\theta\in\Theta} \Pr(\theta \mid \mathcal{D})$.

Even in the general case where $(*)$ does not hold, with access to both the samples θ_i and probabilities $\Pr(\theta_i \mid \mathcal{D})$, we can approximate the optimal solution simply by finding the sample maximum: $\arg\max_{i=1,\ldots,N} \Pr(\theta_i \mid \mathcal{D})$.

... But should we?

One topic in machine learning which has recently garnered attention is *conformal prediction*.[2] The rise in the method's popularity is thanks to its ability to characterize the *uncertainty* of predictions through a confidence interval. Of course, a Bayesian model should naturally allow us to quantify the uncertainty of a "best" model as we are working with distributions.

Question 1.4: Being Uncertain (2 Points)

For the sample mean $\frac{1}{N}\sum_{i=1}^N \theta_i$ and sample maximum $\arg\max_{i=1,\ldots,N} \Pr(\theta_i \mid \mathcal{D})$, discuss how we might (or might not be able to) quantify the uncertainty of the parameter estimate θ given a fixed set of N samples. In other words, describe **two** methods that could establish "error bars" for estimates of θ?

[2] See, for instance: `https://neurips.cc/virtual/2022/invited-talk/55872`

Section 2: Bias and Variance of Regression Models (60 Points)

Let's start from a familiar 1-D regression problem, shown in Figure 2.

The blue line denotes the underlying function that we desire to learn (but do not know):

$$h(x) = 10 \cdot \sin(x) \cdot \exp\left\{-\frac{(x-\mu)^2}{\sigma^2}\right\}, \ x \in [-3, 3] \tag{2.1}$$

For this question we work with $\mu = 0, \sigma^2 = 1$, the shape of $h(x)$ is shown in dark blue.

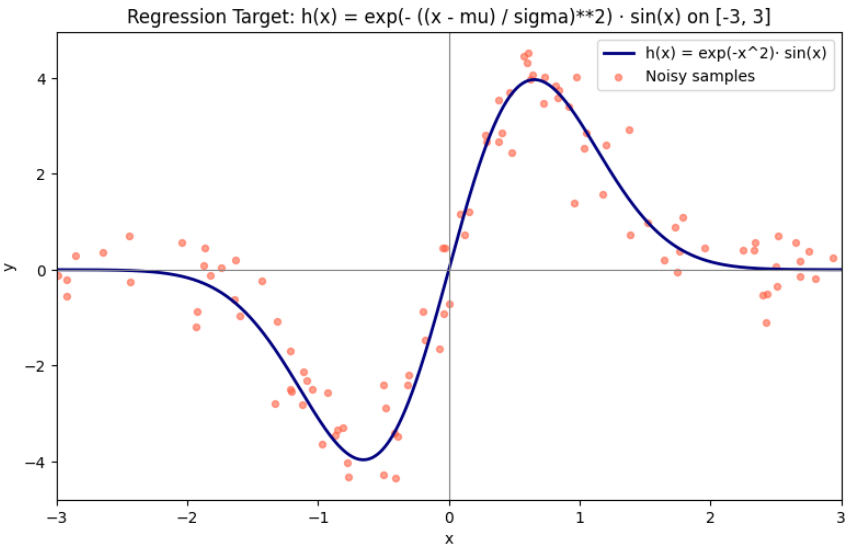

Figure 2: A 1-D regression problem.

We first generate $N_{train} = 100$ data points from $h(x)$ with *iid* Gaussian noise with $noise_std = 0.5$, these points are shown in pink. Throughout this question, we assume that values of x are sampled *iid* from a uniform distribution between $[-3, 3]$.

Question 2.1: Perform regression (5 Points)

Fill in code to fit a linear regression model with an 8-th order polynomial kernel. In particular, you will need to implement `poly_features(x, degree)` that generates a feature matrix `PhiX` given input x and the desired degree, `theta_MLE` that calculates an MLE solution to the weight vector θ, and `predict(x, theta)` that does the prediction.

Question 2.2: Unpack the expressions of bias, variance, and noise (10 Points)

It may help to first review the derivation of Equation (3.41) in the PRML book, which are contained in Section 3.2 and also the section leading up to Equation (1.90). Notice that in the bias-variance-noise decomposition Equation (3.41 – 3.44):

- $h(x)$ refers to the underlying function, i.e. Equation (2.1);
- t is the target value correspond to input x in the (test) dataset, usually assumed to be a noisy version of $h(x)$;
- $y(x; \mathcal{D})$ refers to model predictions made at input point x, when the parameter estimates are obtained from (training) dataset $\mathcal{D}$;
- $\mathbb{E}_{\mathcal{D}}[y(x; \mathcal{D})]$ is a notable new quantity. It represents the *average model prediction* at x, where the expectation is taken over a set of training datasets $\mathcal{D}$. This term is needed in computing both the bias and variance.

Let's denote a test dataset as $\{(x_1, t_1), (x_2, t_2), \ldots, (x_i, t_i) \ldots, (x_M, t_M)\}$, or $\{(x_i, t_i)\}_{i=1}^{M}$ for short. Let's also assume there are K different sets of model parameters $\{\theta_k\}_{k=1}^{K}$, obtained from K corresponding training datasets sampled *iid* from the same distribution as shown in Figure 2.

Let's first expand the noise term using our dataset $\int \text{var}[t|x]p(x)dx = \iint \{h(x) - t\}^2 p(x, t)\, dx\, dt$ – the first part is from PRML (1.90), the second part from (3.37) and (3.44). Note that in the released pdf, (3.44) is missing one integral sign.

Assuming we have D training sets $\{\mathcal{D}_i\}_{i=1}^{D}$ obtained from sampling the ground truth dataset distribution $p(\mathcal{D})$. For this question, please complete the following tasks in order:

(a) In general, it is impossible to obtain analytical solutions for the (bias)^2, variance and noise terms. Explain why this statement is correct.

(b) Is it correct to approximate the double integral in PRML (3.44) as a sum over test data points, as shown in Equation (2.2)? If yes, justify your answer with mathematical reasoning and/or derivations. If no, please write out the correct expression.

$$\text{noise_estimate} = \frac{1}{M} \sum_{i=1}^{M} [h(x_i) - t_i]^2 \tag{2.2}$$

(c) Write out the expression for approximating the (bias)^2 term using our test dataset.

(d) Write out the expression for approximating the variance term using our test dataset.

Question 2.3: Compute (approximated) bias, variance, and noise term (10 Points)

Please generate NUM_DATASET = 50 training datasets, and produce linear regression model weights from each dataset. Generate another set of $M = 1,000$ test data points. With the test dataset, approximate and report

- the noise term;
- the (bias)^2 term;
- and the variance term

using the correct expressions in Question 1.2.

Question 2.4: Expectations and relationships among these error terms (10 Points)

Provide short ($\leq$ 1 line) answers to the following questions.

- What do you expect the noise term should be? and what is the difference between what it *should be* and what was computed in Question 1.3? Let's call this difference ϵ.
- Name *one quantity* that you can change in this regression problem that will likely reduce ϵ.
- Name *one quantity* that you can change in this regression problem that will likely reduce the approximation error in all bias, variance and noise terms.
- What do you expect the three terms computed in Question 1.2 approximately add up to?

Question 2.5: Explore bias vs variance of different models and settings (15 Points)

(a) Now consider different problem settings with NUM_DATASET $= \{5, 10, 20, 40, 80\}$, $N_{\text{train}} \in \{50, 100, 200, 400, 800\}$ and $M = 1000$.

- Fix NUM_DATASETSs to be 20. Vary N_{train}.
- Fix N_{train} to be 200. Vary NUM_DATASETSs.

For both subtasks, create a 2-D scatter plot with five points, whose (x, y)-coordinates are the $(\text{bias})^2$ and variance values. Clearly label the problem setting for all points.

(b) Fit linear regression with different basis functions and regularisations. Create new bias-variance scatter plots with NUM_DATASET $= 10, N_{\text{train}} = 500$ and $M = 1000$ for both tasks below. Comment on your findings.

- Use polynomial basis functions with degrees {4, 8, 12, 16, 32}
- Use {5, 10, 20, 50, 100} Gaussian basis functions of the form:

$$\phi(x) = \exp\left\{-\frac{(x-c)^2}{2d^2}\right\}$$

The centers of Gaussian basis functions c are evenly spaced between -3 and 3, and width $d = 1$.

(c) Implement `theta_MLE_regularised` that adds regularization to the set of regression with polynomial basis functions. Find a suitable value of $\ln \lambda$ (defined in PRML Eq (1.4) and Eq (3.27)) when the degree of the polynomial is 16.

(d) With which degree of the polynomial basis functions does the regularisation benefit the regression the most? Provide an explanation. No need to show the plots.

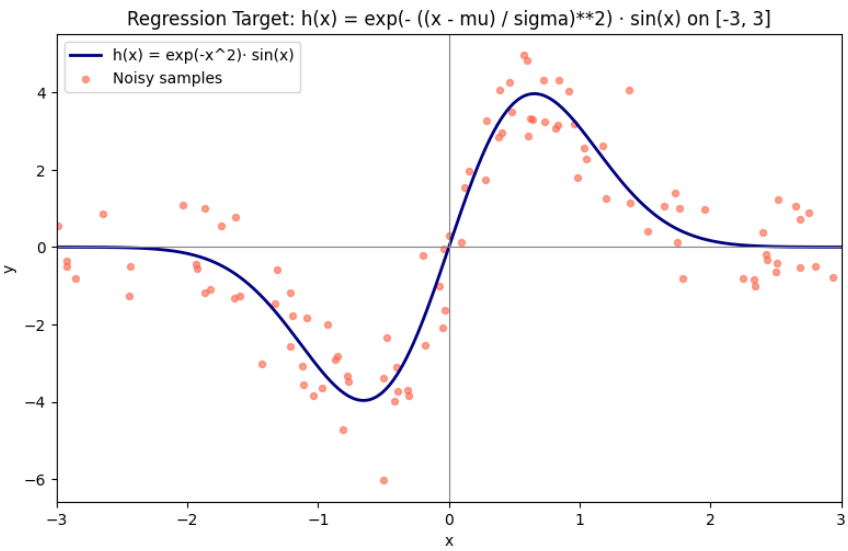

Figure 3: A 1-D regression problem, with lognormal noise (mean=-1 std=0.5 for the underlying Gaussian, and a coin flip to have both positive and negative noise magnitude).

Question 2.6: Regression with non-Gaussian noise (10 Points)

A team of geologists have taken our existing model for this regression task to a newly-formed island in the middle of Pacific ocean. The rock formed by a recent volcanic eruption has abnormal gravity patterns. In particular, geologists thought that the noise follows the log-normal (or lognormal) distribution, a continuous probability distribution of a random variable whose logarithm is normally distributed.

`sample_dataset_lognormal` will generate a regression dataset with the same $h(x)$, but corrupted with lognormal-distributed noise, with a 50-50 chance of the noise being positive or negative in value.

(a) Derive the probability density function (*pdf*) of a log-normal random variable $U = e^{\beta^{-1}Z}$, where $\beta > 0$, Z is distributed according to a standard normal distribution.

(b) The results of (a) can also be written as another form that includes the mean μ and standard deviation σ, as shown below:

$$p(X = x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\ln x - \mu)^2}{2\sigma^2}\right\}$$

Show that it is equivalent to the form derived in (a).

(c) Write out the likelihood function $\mathcal{L}(\theta) = p(y_{1:n}|\phi_{1:n}, \theta)$ of linear regression with lognormal noise, assuming the training set is $\phi_{1:n} = \{\phi(x_1), \phi(x_2), \ldots, \phi(x_n)\} := \{\phi_1, \phi_2, \ldots, \phi_n\}$ with targets $y_{1:n} = \{y_1, y_2, \ldots, y_n\}$, $\mu = -1$ and $\sigma = 0.5$.

(d) Calculate the gradient $\nabla_\theta \ln \mathcal{L}(\theta)$.

Food for thought (these questions are ungraded and you do not need to submit them):
How would you perform parameter estimation on such a model, does the normal equation still apply? For your interests, Part III of The Fundamentals of Heavy Tails book and research papers such as this one https://dl.acm.org/doi/10.1145/3357713.3384329 tackles these problems.

Section 3: Exponential Families (30+5 Points)

For this question, we will explore a unified family of distributions: the *exponential family* (not to be confused with an exponential distribution). The exponential family generalizes many commonly used distribution (Gaussian distribution, Beta distributions, Binomial distributions, *etc.*) and has many interesting properties which are used throughout ML.

First, let us define the exponential family distributions we will be considering in this question.

Definition 1 (Exponential Family[3])**.** Given a function $\boldsymbol{u} : \mathbb{R}^n \to \mathbb{R}^m$, we denote an n-dimensional exponential family distribution as $\textsc{Exp}(\boldsymbol{u}, \boldsymbol{\eta})$, where $\boldsymbol{\eta} \in \mathcal{P} \subset \mathbb{R}^m$ designates the m-dimensional parameters of the distribution within an exponential family[4]. The corresponding densities of the distributions are given by

$$q(\boldsymbol{z} \mid \boldsymbol{\eta}) = \exp(\boldsymbol{\eta}^\top \boldsymbol{u}(\boldsymbol{z}) - \psi(\boldsymbol{\eta})), \tag{3.1}$$

where

$$\psi(\boldsymbol{\eta}) = \log \int_{\mathbb{R}^n} \exp(\boldsymbol{\eta}^\top \boldsymbol{u}(\boldsymbol{z}))\mathrm{d}\boldsymbol{z}. \tag{3.2}$$

The function $\boldsymbol{u}$ is called the sufficient statistics of the exponential family and the function ψ is called the log-partition function of the exponential family. Sometimes, it will be convenient to define $Z(\boldsymbol{\eta}) = \int_{\mathbb{R}^n} \exp(\boldsymbol{\eta}^\top \boldsymbol{u}(\boldsymbol{z}))\, \mathrm{d}\boldsymbol{z} = \exp(\psi(\boldsymbol{\eta}))$ as the normalizer of the distribution.

Remark. We note that the definite integral in Eq. (3.2) is over the domain of $\boldsymbol{z}$. In this assignment, the domain of integration will simply be over $\mathbb{R}^n$; which can be simplified to integrating over $(-\infty, \infty)$ for different coordinates.

Remark. Notice that Definition 1 represent a broad set of special cases of the definition given in the Bishop textbook Eq. (2.194), with $h(\boldsymbol{x}) = 1$ and $g(\boldsymbol{\eta}) = \exp(-\psi(\boldsymbol{\eta}))$. For notational convenience in the tasks specified below, we use this definition.

At this point, it is useful to verify that the Exponential family indeed generalizes distributions we are familiar with.

Question 3.1: Gaussian R.V. as an Exponential Family (10 Points)

Let $\boldsymbol{u}(z) = (z, z^2)$ and $\boldsymbol{\eta} = (\mu/\sigma^2, -1/2\sigma^2)$ define an $(n = 1)$-dimensional exponential family distribution $\textsc{Exp}(\boldsymbol{u}, \boldsymbol{\eta})$ over $\mathbb{R}$.

Verify that $\textsc{Exp}(\boldsymbol{u}, \boldsymbol{\eta})$ is equivalent to the 1-dimensional Gaussian distribution with mean μ and s.t.d. σ.

What is the distribution's normalizer $Z(\boldsymbol{\eta})$?

One interesting aspect of Exponential families is its connections to loss functions in ML one typically finds. The simplest connection comes from 'equating' a loss function minimization procedure to the MLE of an exponential family, *i.e.*,

$$\arg\min \text{“A loss function } \ell\text{”} = \arg\max \log q(\boldsymbol{z} \mid \boldsymbol{\eta}). \tag{3.3}$$

Of course, to do this, one needs to find the correct matching between losses and exponential family. To do so, let us introduce some data.

Assumption: Let us assume that we have data $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$ with features $\boldsymbol{x}_i \in \mathbb{R}^n$ and corresponding labels / targets $y_i \in \mathbb{R}$.

[3] This is actually a simplified version of an exponential family. A more general version can be defined, *i.e.*, non-unit base measure.

[4] The exponential family for a fixed $\boldsymbol{u}$ is the set of distributions given by $\mathcal{M}_{\boldsymbol{u}} = \{\textsc{Exp}(\boldsymbol{u}, \boldsymbol{\eta}) \mid \boldsymbol{\eta} \in \mathcal{P}\}$. Two exponential family distributions share the same exponential family if and only if their $\boldsymbol{u}$'s are the same.

Question 3.2: Exponential Families and Losses (5 Points)

Let $f(\boldsymbol{x};\boldsymbol{w}) = \boldsymbol{w}^\top \boldsymbol{x}$ with weights $\boldsymbol{w} \in \mathbb{R}^n$. Let us define a conditional random variable using a 1-dimensional exponential family, such that

$$y \mid \boldsymbol{x} \sim \textsc{Exp}(\boldsymbol{u}, \boldsymbol{\eta}), \tag{3.4}$$

where $\boldsymbol{u}(y) = (y, y^2)$ and $\boldsymbol{\eta} = \boldsymbol{\eta}(\boldsymbol{x}) = (f(\boldsymbol{x};\boldsymbol{w}), -1/2)$.
That is, as per Definition 1, we are defining the following conditional p.d.f. such that:

$$p(y \mid \boldsymbol{x};\boldsymbol{w}) = q(y \mid \boldsymbol{\eta}(\boldsymbol{x})) = \exp(\boldsymbol{\eta}^\top(\boldsymbol{x})\boldsymbol{u}(y) - \psi(\boldsymbol{\eta}(\boldsymbol{x}))). \tag{3.5}$$

Show that

$$\arg\max_{\boldsymbol{w}\in\mathbb{R}^n} \sum_{i=1}^{N} \log p(y_i \mid \boldsymbol{x}_i;\boldsymbol{w}) = \arg\min_{\boldsymbol{w}\in\mathbb{R}^n} \sum_{i=1}^{N} (y_i - f(\boldsymbol{x}_i;\boldsymbol{w}))^2. \tag{3.6}$$

Remark. One might notice that in Question 3.2, the functional form of f and its corresponding weights $\boldsymbol{w}$ does not effect the maximization / minimization.

In the general case, optimizing either Eq. (3.6) analytically ('by hand') can be difficult, *e.g.*, when f is a neural network. As such, it is useful to consider the gradients of the log-likelihood function.

Question 3.3: Parameterizing an Exponential Families (15 Points)

Let $f(\boldsymbol{x};\boldsymbol{\theta})$ be some mapping from feature $\boldsymbol{x}$ parameterized by $\boldsymbol{\theta}$. Let us define a conditional random variable using an exponential family, such that

$$\boldsymbol{y} \mid \boldsymbol{x} \sim \textsc{Exp}(\boldsymbol{u}, \boldsymbol{\eta}), \tag{3.7}$$

where $\boldsymbol{\eta} = f(\boldsymbol{x};\boldsymbol{\theta})$ for any function $\boldsymbol{u} : \mathbb{R} \to \mathbb{R}$ (we hide the dependence of $\boldsymbol{x}$ in $\boldsymbol{\eta}$ to avoid notation clutter). We also define $p(\boldsymbol{y} \mid \boldsymbol{x};\boldsymbol{\theta})$ similar to Eq. (3.5).
Show that

1. Show that
$$\left.\frac{\partial \psi(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}}\right|_{\boldsymbol{\eta}=f(\boldsymbol{x};\boldsymbol{\theta})} = \mathbb{E}_{p(\boldsymbol{y}\mid\boldsymbol{x})}[\boldsymbol{u}(y)]. \tag{3.8}$$

2. Show that
$$\frac{\partial \log p(\boldsymbol{y} \mid \boldsymbol{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \left(\boldsymbol{u}(\boldsymbol{y}) - \mathbb{E}_{p(\boldsymbol{y}'\mid\boldsymbol{x})}[\boldsymbol{u}(\boldsymbol{y}')]\right)^\top \left(\frac{\partial f(\boldsymbol{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right). \tag{3.9}$$

Eq. (3.9) allows us to use *gradient ascent* (confer, gradient descent) to iteratively update the parameters of our model. Of course, we can practically use Eq. (3.9) by using auto-differentiation packages in Python. However, one will need to do some tricks to make the computation efficient.

Question 3.4 [COMP4670/8600]: Implementing Gradient Ascent (+5 Points)

In '`exponential_family.py`' complete the following functions:

- `grad_psi_wrt_eta` (Eq. (3.8));
- `log_likelihood_grad` (Eq. (3.9));
- `update`;

and thus train a neural network in `display.ipynb` for the MNIST dataset (you may need to install some dependency packages, e.g. `torch`, `texlive`, etc., to run successfully).
You will need to use `from torch.autograd.functional import jacobian, vjp`.

Hint: You may want take a look at `framework/exp_fam_model.py` (or the copied comment in `exponential_family.py`).

You will not be graded on optimality of the neural network's performance, only the correctness of implemented functions.